Application of Graphene-based 2D Materials in Energy Storage Devices

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by

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1. ABSTRACT

The rapid growth of renewable energy systems, electric vehicles, portable electronics is driving the demand for advanced energy storage systems with high capacity, fast charge-discharge rates and long cycle life.

Graphene, an allotrope of Carbon possessing incredible properties such as exceptional surface area, electrical as well as thermal conductivity and mechanical strength has the potential to disrupt the traditional energy storage scenario by enhancing the performance of Lithium-ion, Sodium-ion batteries. Graphene also has potential to significantly improve the performance of Electric Double-Layer Capacitors (EDLCs), Pseudocapacitors and Hybrid capacitors because of its theoretical specific surface area of up to 2630 m^2/g.

Additionally, Graphene and its oxide derivatives have been used in the fabrication of supercapacitors and supercapacitor electrodes. These composite supercapacitors electrodes show excellent capacitance and can be used to develop capacitors that are better than traditional counterparts on several parameters.

This report discusses the application of graphene and its derivative compounds as well as composites in the energy storage space. The challenges and prospects are also discussed.

2. INTRODUCTION

Graphene is a very revolutionary material that is capable of fundamentally changing the energy storage systems landscape. It is a single-atom thick sheet of carbon atoms that are arranged in a hexagonal lattice. It has extraordinary physical and chemical properties that can be used in developing next-generation energy storage systems. These include superior electrical conductivity, very high tensile strength and a very high theoretical surface area. These properties enable graphene-based energy storage systems to surpass the capabilities of traditional energy storage systems. We shall discuss the factors that make graphene suitable for energy storage.

The two-dimensional planar structure of graphene offers several advantages over conventional materials.

A. Enhanced Electron and Ion Transport

The hexagonal structure of graphene is very uniform. This allows rapid passage of electrons across the material. The ultra-high electrical conductivity enables fast charge and discharge rates. This directly translates into high power density for devices like supercapacitors and batteries. The large surface area also allows efficient transportation and adsorption of ions. This greatly improves the stability and reliability of batteries.

B. Maximizing Surface Area and Energy Density

The theoretical surface area of graphene is enormous. This property is very important in Electrochemical Double Layer Capacitors (ELDCs). A large surface area allows larger charge storage resulting in higher capacitance and higher energy density compared to traditional materials like activated carbon. Moreover, the 2D structure means most of the storage space will be utilized.

C. Versatility as an Electroactive Material

An electroactive material is a material whose physical properties can be modified by the effect of an electric field. Graphene is an excellent electroactive material. It can be used in ELDCs because of its high surface area. Graphene can also be used as a highly conductive mix/additive to enhance the performance of batteries and supercapacitors. It can also be used as a host material for various metal oxides, sulfides or polymers to

create graphene-derivative composites which are vital for enhancing the performance and stability of batteries and supercapacitors.

The application domain of graphene-based 2D materials in energy storage is vast but, this report focuses on two major technologies: Lithium-ion Batteries and Supercapacitors.

1. **Lithium-ion Batteries**

   Graphene and its derivatives like Reduced Graphene Oxide (rGO), are instrumental in addressing the current limitations of lithium-ion batteries by improving the rate capability, cycle life and energy density.

   Graphene is also an excellent alternative to traditional graphite anodes. Its high electrical conductivity drastically improves charge-discharge rate enabling faster charging. Furthermore, graphene can also be used as a conductive and flexible buffer to stabilize the high-capacity anode materials. Graphene can also mitigate the mechanical stress by preventing electrode pulverization resulting in extending cycle life.

2. **Supercapacitors**

   Supercapacitors are known for their high-power density and long cycle life. These properties can be further enhanced by the incorporation of graphene. Graphene is a very versatile material therefore, it can be used in both EDLCs and Hybrid/Pseudocapacitors.

   In EDLCs, graphene can be used to create electrodes with ultrafast ion response times and high capacitance. Graphene has very low electrical resistance which directly translates to higher power density.

   In Hybrid and Pseudocapacitors, Graphene is often composited with pseudocapacitive materials like RuO2 which store energy in fast reversible Faradic reactions.

In summary, Graphene is a foundational component in next-generation energy storage devices. Its unique properties allows it to address the limitations of current technologies and push the boundaries of innovation in batteries and supercapacitors across a multitude of applications.

1. ENERGY STORAGE DEVICES: OVERVIEW

The technological advancement of humanity is owed to the advancements in energy and storage technology. To progress further, we need to develop cleaner and more efficient ways of storing energy. It is therefore important to know what technologies we use today.

We shall discuss three methods of energy storage that are extensively used in the current tech scenario.

1. LITHIUM-ION BATTERIES

Lithium-ion Batteries are the widespread rechargeable battery technology. They have very high energy density. They function by the movement of Li^+ ions between a cathode and anode, typically graphite through a non-aqueous electrolyte.

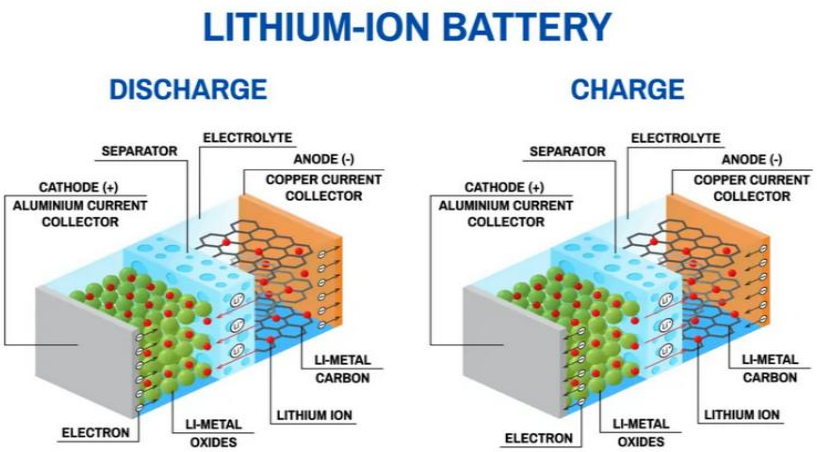

image 3.1.1: lithium-ion battery

The above image shows the charge and discharge process of a typical Lithium-ion battery. The cathode is typically made of high nickel layered oxides. The anode is made of graphite or silicon-carbon composites. A separator is added to prevent short circuits.

Lithium-ion batteries come in a variety of shapes and sizes for different applications. Some examples include cylindrical, prismatic and pouch cell formats.

image 3.1.2 : kinds of lithium-ion battery

The technical specifications of Lithium-ion batteries are as follows:

Feature	Typical Range
Specific Energy	120 - 270 Wh/kg
Specific Power	250 - 350 W/kg
Cycle Life	500 - 5,000 cycles
Nominal Voltage	~3.3 - 3.8 V per cell
Efficiency	~85 - 95%

table 3.1.1: technical specifications of lithium-ion battery

However, high costs, safety concerns and performance degradation over time remain prominent drawbacks of such batteries. Moreover, recycling these batteries is another major challenge.

2. SUPERCAPACITORS

Supercapacitors are devices that store energy electrostatically. They do this by the adsorption of ions onto electrode materials that have a very high surface area.

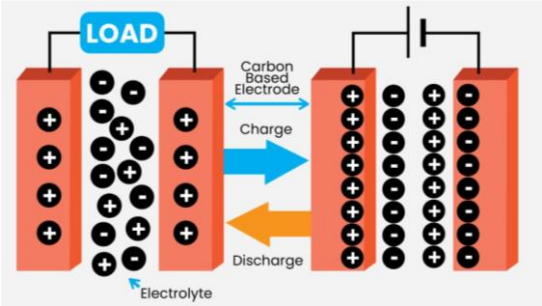

image 3.2.1: working of an ELDC capacitor

The picture shows how an EDLC (Electric Double Layer Capacitor) supercapacitor works. During charge, the positive and negative ions in the electrolyte are attracted to the oppositely charged electrodes that are carbon based. During discharge, these ions are released, releasing stored energy. All of this happens in a fraction of a second allowing rapid energy storage and release.

image 3.2.2: various kinds of capacitors

Supercapacitors are developed using high surface area carbon materials such as graphene, activated carbon and carbon nanotubes. Aqueous electrolytes that are used include KOH, H2SO4, NA2SO4. These days gel type electrolytes are also used such as polyvinyl alcohol (PVA). Introducing metal oxides or conductive polymers can increase the capacity of these capacitors.

Supercapacitors are used in applications requiring near instant or rapid release of energy. Therefore, they are used in actuators in wind turbines, cranes, lifts, sensors, LED flashes, portable electronics, circuits and others.

The general specifications of a supercapacitor are as follows:

Feature	**Typical Range**
Specific Energy	<10 Wh/kg
Specific Power	Up to 10,000 W/kg
Cycle Life	Hundreds of thousands to over a million cycles

Charge/Discharge Time	Seconds to Minutes
Efficiency	~95 - 98%

table 3.2.1: technical specifications of a general supercapacitor

2. GRAPHENE AS A 2D MATERIAL IN ENERGY STORAGE

Owing to their unique structures and properties, carbon allotropes have emerged as potential candidates for a variety of applications in material science. Graphene presents a hexagonal 2D structure which has exceptional mechanical, thermal and electrical properties. It also presents significant potential for enhancing composite materials across various applications.

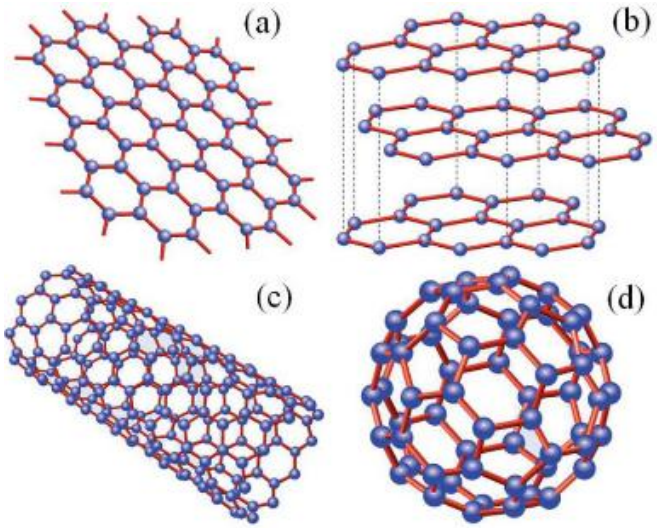

image 4.1: graphene and its related structures

The image above represents graphene and its related structures. (a) Graphene structure 2D hexagonal lattice of carbons (b) A-B-A stacked graphene layers. (c) Rolled up cylinder of graphene. (d) C60 is a molecule formed by balling up graphene into a sphere resulting in the formation of pentagons and hexagons in its structure. This molecule is also known as fullerene.

Graphene is an excellent additive to nanocomposites. Graphene nanocomposites have become very popular as they can enhance properties of materials with very low concentrations, normally 1% - 3% by weight.

Graphene has extraordinary mechanical strength; it is around 200 times stronger than steel while being very lightweight. Its high thermal conductivity allows it to dissipate heat effectively. This makes it ideal for electronic and energy storage applications. Moreover, graphene has excellent electrical conductivity which tremendously increases the performance of conductive composites. The physical properties of graphene are given below:

Physical Property	Value
Specific surface	2600 m^2/g
Young's module	0.5 TPa
Breaking stress	42 N m
Thermal conductivity	5000 W/m K
Electric conductivity	0.96×10^8 Ohms·m^{-1}
Refractive index at 670 nm	3.135
Absorption coefficient at 670 nm	0.897

table 4.1: technical specifications of a general supercapacitor

One of the most promising areas for graphene is energy storage. Especially in batteries and supercapacitor technologies. Integrating graphene into electrode materials has shown significant improvement in these aspects. For example, graphene oxide can increase the surface area of electrodes allowing faster charge and discharge cycles. Compositing graphene with other substances has led to the development of hybrid supercapacitors that offer very high energy density while maintaining fast charging and long-term durability. Some other applications of graphene in energy storage include:

1. Photovoltaic Cell
2. Conductive Ink
3. Paper-based Electrode
4. Battery Cathode
5. Battery Anode
6. Capacitors

3. BATTERIES AND GRAPHENE

A battery is a device that converts chemical energy into electrical energy. The technology that we use today is unimaginable without high performance batteries. Therefore, a lightweight secondary battery capable of high-energy storage and small charging time is necessary for modern electrical gadgets.

The modern battery scenario is dominated by Lithium-ion batteries owing to its unmatched power density and commercial availability.

The speed of transportation of Li ions between the electrodes through the electrolyte determines the power of a Lithium-ion battery. Thus, a large surface area with high electrical conductivity is needed for making a high-power Lithium-ion battery. Guess which material fulfils both of those requirements? Graphene!

These batteries use graphite and expanded graphite as anodes. An electrochemical comparison between graphite, expanded graphite and graphene nanosheet shows that the reversible capabilities of graphene nanosheets are almost double than expanded graphite electrodes and three times as much as graphite electrodes.

We shall explore the application of graphene in these batteries:

1. **APPLICATION OF GRAPHENE AS CATHODE MATERIALS IN LITHIUM-ION BATTERIES:**

   Current cathode materials such as $LiFePo_4$ (LFP), suffer from poor conductivity and low Li^+ mobility. Integrating graphene with LFP can improve its performance.

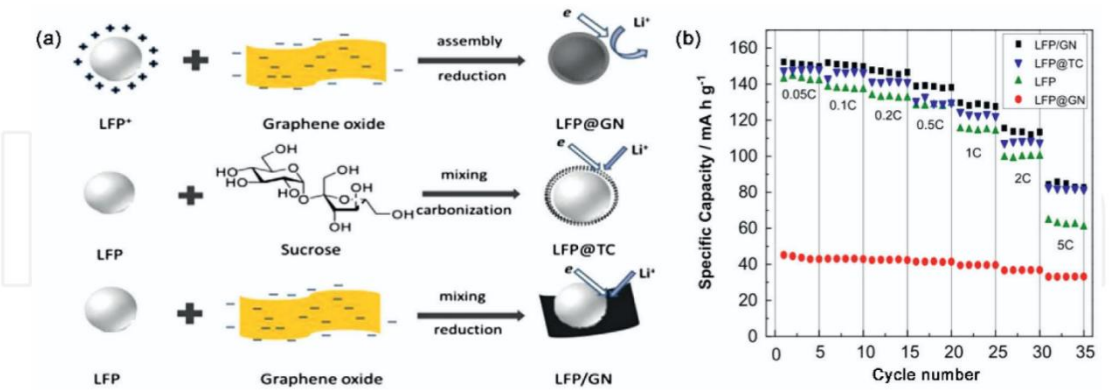

image 5.1: LFP/C/G composite synthesis

This image (a) shows LFP/C/G composites that were synthesized using a high-energy ball milling assisted rheological phase method. These formed 3D “ball-in-chip” and “ball-on-chip” conductive network structures which rich mesopores. These structures enhance electron and ion transportation, increasing battery performance.

The second image (b) shows that with only 3% by weight graphene content, the batteries achieved high initial discharge capacities and excellent cycling stability showing only 8% capacity loss after 500 cycles at 10C.

This means that graphene, at even low concentrations, can significantly improve charge capacity and stability of lithium-ion batteries.

2. APPLICATION OF GRAPHENE AS ANODE MATERIALS IN LITHIUM-ION BATTERIES:

Traditional graphite anodes have low specific capacity. The formation of lithium dendrites can pose a massive safety risk. Integrating graphene in anodes can mitigate this safety risk.

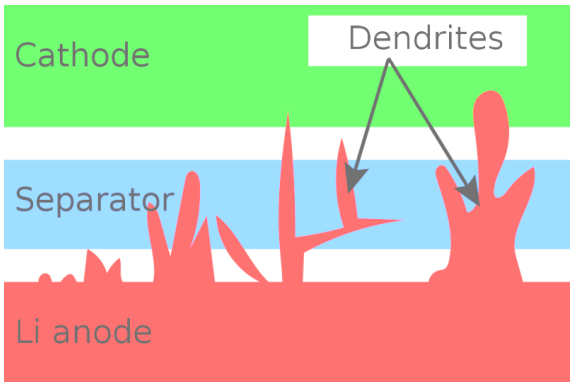

image 5.2.1: Li dendrite formation

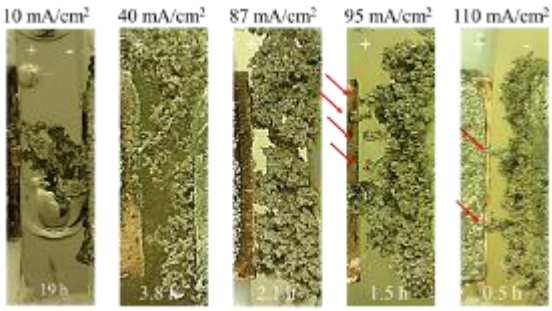

image 5.2.2: Li dendrite formation under digital optical microscope

The images above show the formation of lithium dendrites that can cause short-circuits posing a massive safety risk.

Graphene based carbon materials can host more lithium by forming LiC_3 with a theoretical capacity of 744 mAh g^{-1} , double that of graphite. Graphene coatings also improve electrochemical performance by enhancing conductivity, increase lithium storage and preventing electrode volume interactions.

For example, Si@Bgra anodes made by Chemical Vapor Deposition (CVD) achieved 2500 mAh g^{-1} at 3 A g^{-1} with 85% capacity retention after 1000 cycles.

Therefore, graphene's excellent electrochemical favoring properties can significantly increase the performance and stability of of lithium-ion battery anodes.

4. SUPERCAPACITORS AND GRAPHENE

In an ordinary capacitor, the charge is stored electrostatically (on metal plates separated by a dielectric). Ordinary capacitors have relatively low capacitance which is in the range of a few microfarads to millifarads. They are widely used in signal filtering, timing circuitry and electronics.

Supercapacitors can store charge electrostatically and electrochemically. Their capacitance can reach hundreds to thousands of Farads. Much higher than ordinary capacitors. They are widely used in hybrid vehicles, backup power and renewable energy systems.

Graphene can significantly enhance the capacitance and performance of these supercapacitors. Graphene is primarily used in electrodes. Usually after combining with other materials such as manganese dioxide, ruthenium oxide.

This greatly improves energy density, power density, cycle life and allows the fabrication of flexible and lightweight storage devices.

1. APPLICATION OF GRAPHENE BASED SUPERCAPACITORS

a. Electric and Hybrid Vehicles

Graphene based supercapacitors play a major role in regenerative braking, torque delivery and load management. This is because graphene-based supercapacitors can do rapid charge transfers and have super-high power densities that exceed 10kW/Kg.

Graphene-hybrid electrodes improve energy density (20-100Wh/Kg) without compromising cycle stability and reliability. This makes them ideal for next-generation EVs.

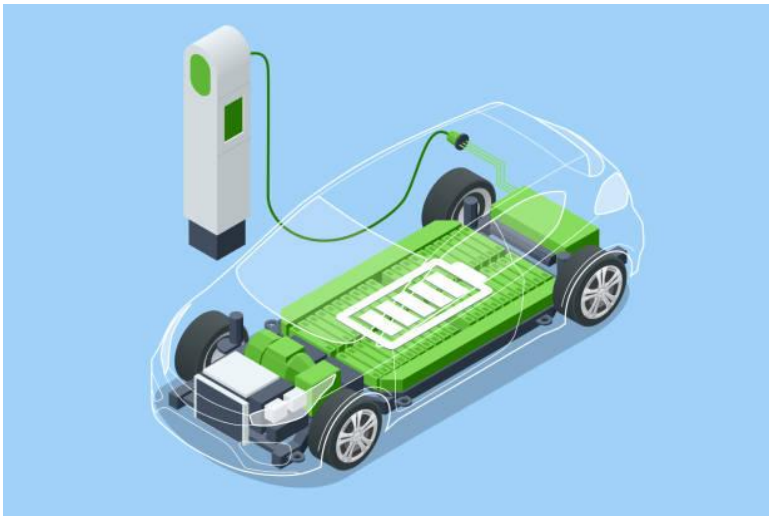

image 6.1.1: EV energy system

b. Portable and Wearable Electronics

Graphene-based solid-state supercapacitors using gel electrolytes like PVA/H3PO4 demonstrate high mechanical endurance and capacitance under stress and strain. These devices can be printed or laminated onto polymers or textiles allowing their use in flexible skins, displays and foldable electronics. This combination of high power density and mechanical reliability allows stable operation under repeated stress.

image 6.1.2: wearable health technology

c. Renewable Energy Systems

Graphene-based supercapacitors are integrated into photovoltaic and wind energy systems for energy buffering and load management. The ultrafast charging and discharging allow them to manage short term fluctuations. These supercapacitors can work in a variety of extreme environment temperature ranges from -40 C to 80 C. This makes them suitable for off-grid power storage, microgrids and energy harvesting systems.

image 6.1.3: renewable energy systems

d. Internet of Things (IoT) Devices

The ultralow power consumption of graphene supercapacitors make them ideal for IoT nodes, wireless sensors and edge computing devices. These use a very small variety of capacitors called microscale graphene-based micro-supercapacitors (MSCs). MSCs have extremely fast response times (<100 ms), high capacitance (~25mF/cm^2) and excellent cycle stability (>10^5 cycles) making them a crucial component in self-powered smart devices.

image 6.1.4: IoT representation

5. CHALLENGES AND FUTURE PROSPECTS

As futuristic and promising graphene in energy storage is. It has challenges. We shall discuss a few below.

1. **Large Scale Production Issues and Cost**

   Producing graphene-electrodes and composites on an industrial scale remains difficult and extremely expensive. The current methods such as CVD and mechanical exfoliation lack scalability.

2. **Structural Stability and Material Degradation**

   During repeated charge and discharge cycles, graphene electrodes experience restacking and agglomeration. Basically, graphene sheets clump together due to weak van der Waal's forces limiting their potential applications.

3. **Safety and Environmental Impact**

   The chemicals and compounds used in the production of graphene are not exactly eco-friendly. They pose environmental and health risks. Moreover, biocompatibility and recycling processes of graphene are yet to be fully understood.

4. **Integration Challenges**

   Developing metal oxide and graphene composites are challenging because of element compatibility. Poor compatibility results in weak bonding, poor ion transmission across interface and degradation in performance.

The prospects of graphene are as follows.

1. **Three-Dimensional Architectures**

   The development of 3D graphene can prevent restacking and clumping issues of graphene. Developing porous 3D graphene structures allow high load capabilities while maintaining fast charge transfers.

2. **Commercialization and Standardization**

For graphene-based energy storage systems, cost and development are the two biggest obstacles. To overcome them, we need to develop cost-effective standard synthesis processes that can be scaled at an industrial level.

3. **Sustainable and Green Approaches**

Eco-friendly graphene synthesis using biomass, electrochemical exfoliation and natural methods is an emerging field. Developing such methods reduce environmental hazards and lowers production costs making graphene technology much more sustainable.

6. CONCLUSION

Graphene is a two-dimensional structure of carbon. It has captured the interest of scientists and engineers because of its incredible properties. It is often regarded as a “wonder material”. Its unrivaled electrical and thermal properties as well as mechanical properties make it one of the most sought-after materials in next-generation technologies. At the forefront is energy storage technologies. Graphene is a game changer in this domain. Its integration in batteries and supercapacitors have resulted in batteries that can last longer, store more energy and are much more reliable than traditional counterparts.

The future of graphene is exceptionally bright, but it is not without its challenges. Continued research and innovation are needed to develop sustainable and scalable synthesis methods. The development of next-generation energy storage devices with graphene will propel our technological future.

In conclusion, graphene is more than just a material. It is the foundational platform that is pushing the limits of energy storage devices. The current innovation and breakthroughs in energy storage technology all point to a future where devices are faster, lighter and more durable.

The success of this future depends on overcoming remaining challenges on large scale, cost-effective manufacturing and standardization of synthesis methodologies. The scientific evidence is clear: graphene-based energy devices are not only an improvement in our current technology but, also the prerequisite for the next generation sustainable technological future.

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